

# Hidden Gems Hotels Recommender

Final Capstone Report

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## Executive Summary

The Hidden Gems Hotels Recommender is a content-based hotel recommendation system designed to surface high-quality but relatively underappreciated hotels in major European cities. Instead of ranking hotels only by popularity or review volume, the project combines guest ratings, aspect-based sentiment, quiet/crowded review signals, relative distance from city center, and user preference matching.

The final notebook results were generated using a 100,000-review development sample for efficient local experimentation. The Streamlit application supports 100k, 200k, and full-dataset processing modes. This report clearly separates the 100k reporting sample from the full-dataset capability of the application.

- The recommender separates baseline hotel quality from user-specific preference adjustments.
- Paris showed the highest average top-hotel hidden gem score among the evaluated cities in the reporting sample.
- Preference settings meaningfully changed results: 7 of the top 10 Paris recommendations changed between quiet and lively traveler scenarios.
- The completed Streamlit app provides city selection, preference controls, ranking tables, map visualization, and explainability metrics.

## Business Problem

Hotel discovery platforms often favor highly reviewed, heavily marketed, or chain properties. This can make smaller or less-publicized hotels difficult to discover, even when guests consistently rate them highly. The stakeholder problem is therefore not simply finding the most popular hotel, but identifying properties that combine strong guest experience signals with lower relative visibility.

This project answers the question: which hotels provide excellent guest experiences while remaining relatively overlooked compared with competitors in the same city?

## Dataset

The project uses the Kaggle 515K Hotel Reviews Data in Europe dataset. It includes hotel names, reviewer scores, positive and negative review text, hotel coordinates, and total review counts for hotels in Amsterdam, Barcelona, London, Milan, Paris, and Vienna.

| Item                                  | Description                                                      |
|---------------------------------------|------------------------------------------------------------------|
| Reporting sample                      | 100,000 reviews used for notebook development and report figures |
| Hotel-level records in exported stats | 1,476 hotels                                                     |
| Cities covered                        | Amsterdam, Barcelona, London, Milan, Paris, Vienna               |
| Application processing options        | 100k reviews, 200k reviews, full dataset                         |

# Methodology

## Data Preparation

Reviews were cleaned and aggregated from the review level to the hotel level. The city field uses Province\_Name renamed to Metro\_Area in the notebook workflow, rather than parsing Hotel\_Address. Hotel-level aggregation produced average rating, review-count features, aspect sentiment features, quiet/crowded features, and distance metrics.

## Aspect-Based Sentiment Analysis

Instead of treating each review as a single sentiment value, the notebook applies aspect-based sentiment analysis. Review sentences are split, matched against aspect-specific keyword sets, scored using TextBlob polarity, and aggregated to the hotel level.

- Aspects: noise, cleanliness, staff, location, and value.
- Generated features: noise\_sentiment, cleanliness\_sentiment, staff\_sentiment, location\_sentiment, and value\_sentiment.

## Quiet vs. Crowded NLP Features

Additional NLP features capture whether review language describes hotels as quiet, calm, relaxing, noisy, busy, or crowded. These features are especially important because the application supports quiet/calm and lively/crowded traveler preferences.

## Geographic Features

Hotel coordinates were compared with fixed city-center coordinates using the Haversine formula. Distance was then transformed into several relative features so that users can prefer central, balanced, or farther-from-center locations.

## Hidden Gem Scoring

The baseline hidden-gem score combines hotel quality and review-derived features while applying a small penalty for high review volume. This creates a ranking that rewards strong guest experience while reducing the tendency to over-promote already-popular properties.

| Feature               | Weight |
|-----------------------|--------|
| avg_rating            | 0.30   |
| cleanliness_sentiment | 0.15   |
| staff_sentiment       | 0.15   |
| location_sentiment    | 0.10   |
| value_sentiment       | 0.10   |
| noise_sentiment       | 0.10   |
| crowded_negative_rate | -0.05  |
| log_review_count      | -0.05  |

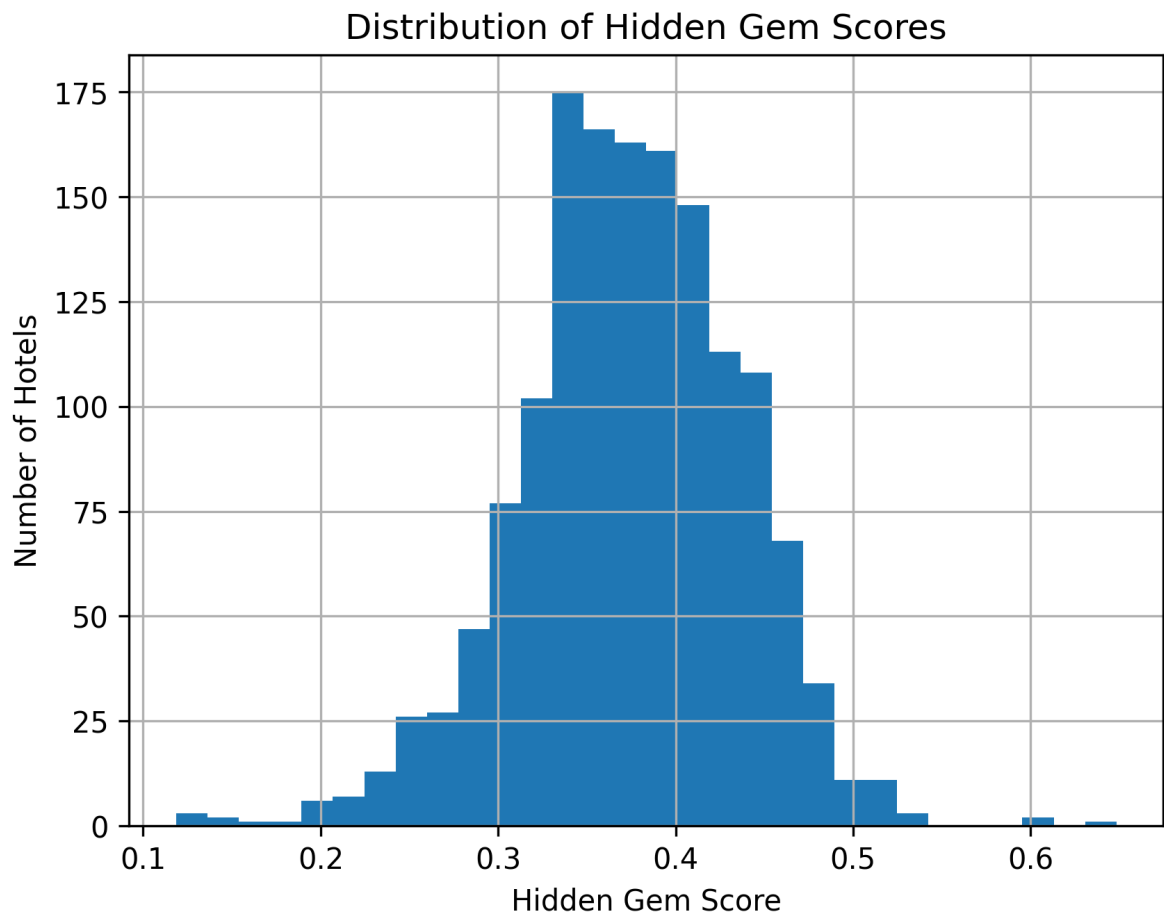
## Personalized Recommendation Engine

The final recommender intentionally separates hidden\_score\_base from user preference matching. Preferences adjust rankings using calm quality, crowd match, and distance match scores. This keeps

the baseline quality logic interpretable while allowing different traveler profiles to receive different recommendations.

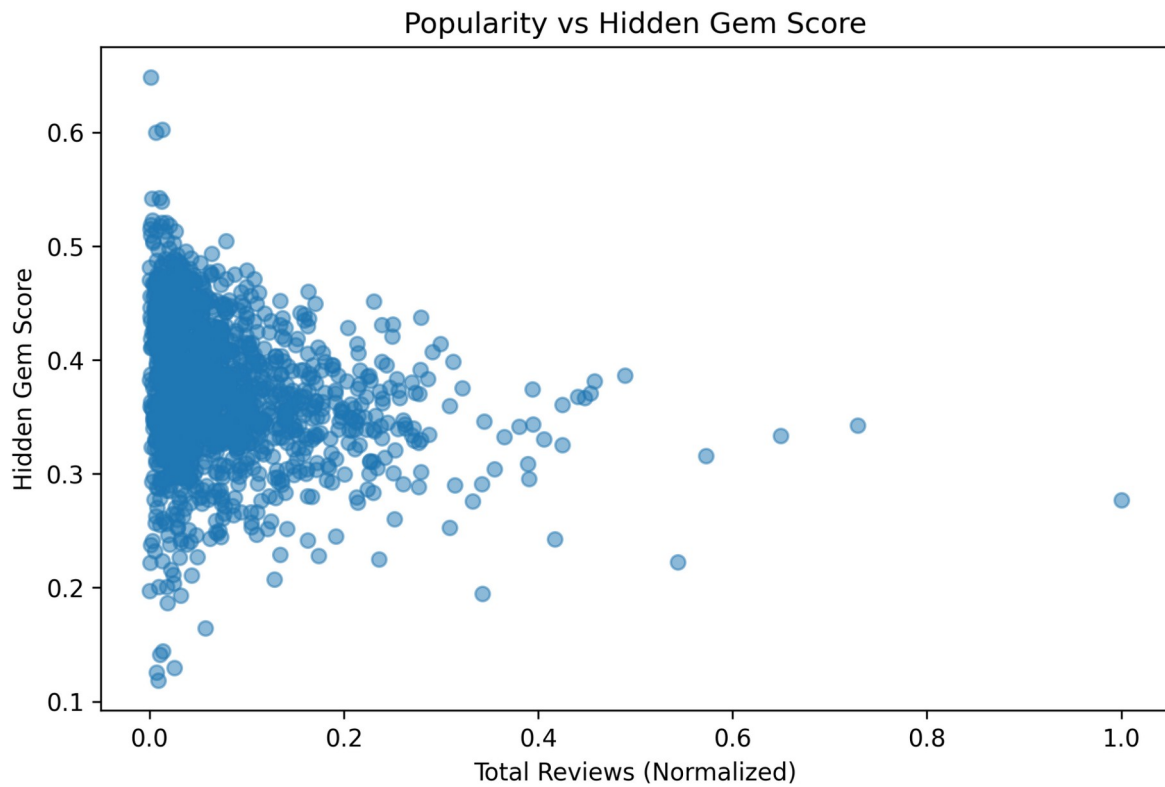
## Results

The results section focuses on a small number of figures that support the project data story: hidden gem scores are measurable, popularity is not the same as quality, city-level opportunities differ, user preferences change rankings, and the analysis is delivered through a working app.



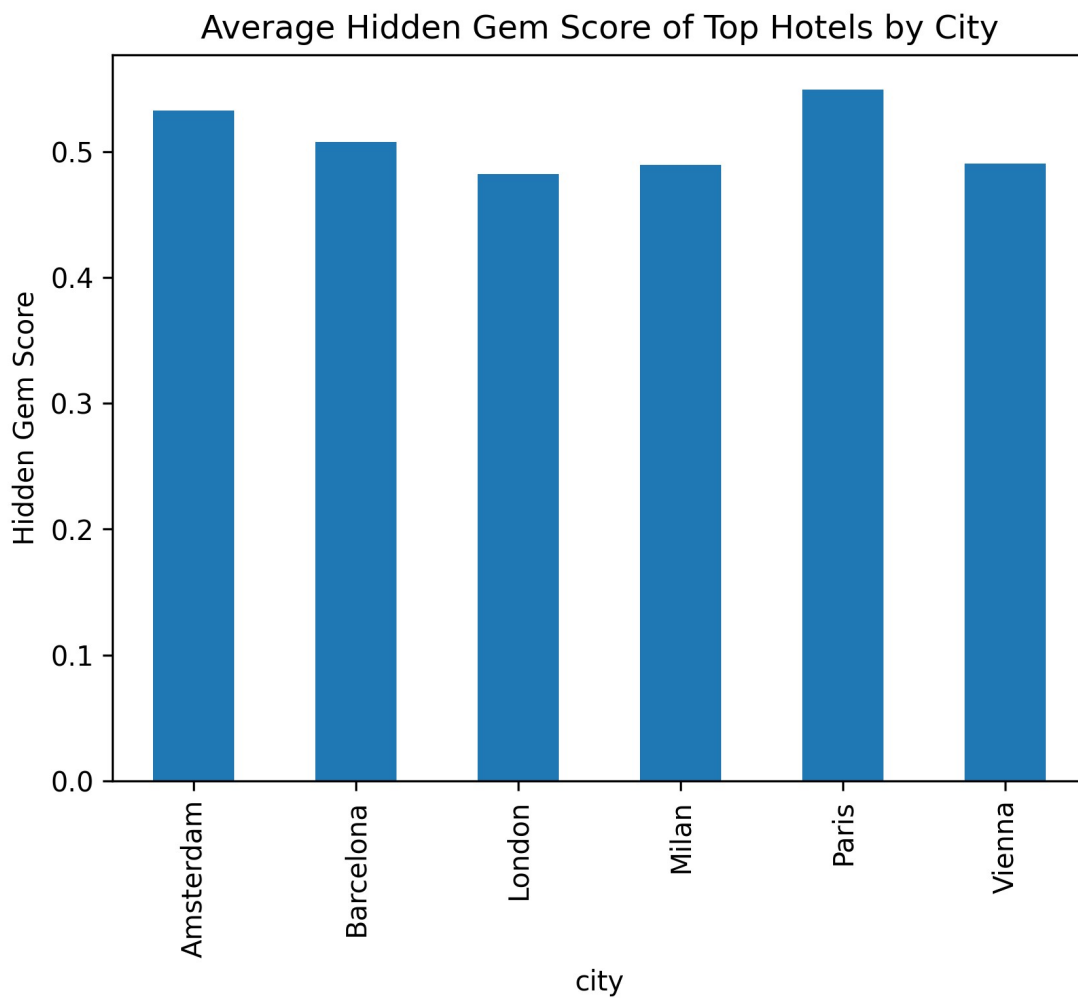
*Figure 1. Distribution of hidden\_score\_base values in the 100k reporting sample. Scores cluster around the midrange, with a smaller number of high-scoring hotel outliers.*

Figure 1 shows that most hotels fall within a relatively compact hidden-gem score range, while a smaller set of hotels score meaningfully higher. This supports using the score as a ranking feature rather than treating all hotels as equivalent once they pass a basic rating threshold.



*Figure 2. Popularity vs. hidden gem score. Total reviews are normalized. The figure shows that higher popularity does not guarantee a higher hidden-gem score.*

Figure 2 supports the central project hypothesis: review volume and hidden-gem quality are not identical. Many hotels with lower normalized review counts still achieve strong hidden-gem scores, while highly reviewed hotels are not always the highest-scoring properties.



*Figure 3. Average hidden-gem score of top hotels by city. Paris has the highest average among top hotels in this sample.*

Figure 3 compares the average hidden-gem score for top hotels by city. The pattern suggests that hidden-gem opportunity differs across markets, with Paris standing out in the reporting sample.

## Impact of User Preferences on Paris Hotel Recommendations

| Rank | Quiet Traveler                        | Lively Traveler                |
|------|---------------------------------------|--------------------------------|
| 1    | H tel Le Walt                         | Hotel Le Mareuil               |
| 2    | Mercure Paris Gobelins Place d'Italie | Splendide Royal Paris          |
| 3    | Hotel Tour d'Auvergne Opera           | Hotel Whistler                 |
| 4    | H tel Baume                           | H tel Le Walt                  |
| 5    | Hotel Villa Lafayette Paris IX        | Shangri La Hotel Paris         |
| 6    | H tel Champs Elyses Plaza             | Lyric H tel Paris              |
| 7    | Mandarin Oriental Paris               | H tel Saint Marc               |
| 8    | H tel Amastan Paris                   | Hotel Villa Lafayette Paris IX |
| 9    | Victoire Germain                      | Mandarin Oriental Paris        |
| 10   | H tel Arvor Saint Georges             | H tel Monna Lisa Champs Elyses |

*7 of the top 10 recommendations changed when traveler preferences changed.*

***Figure 4. Impact of user preferences on Paris recommendations. Seven of the top ten recommendations changed when traveler preferences changed.***

Figure 4 demonstrates that the recommender is not simply returning one static list. When preference settings shift from quiet/calm to lively/crowded, most of the top ten recommendations change, showing that personalization meaningfully affects ranking output.

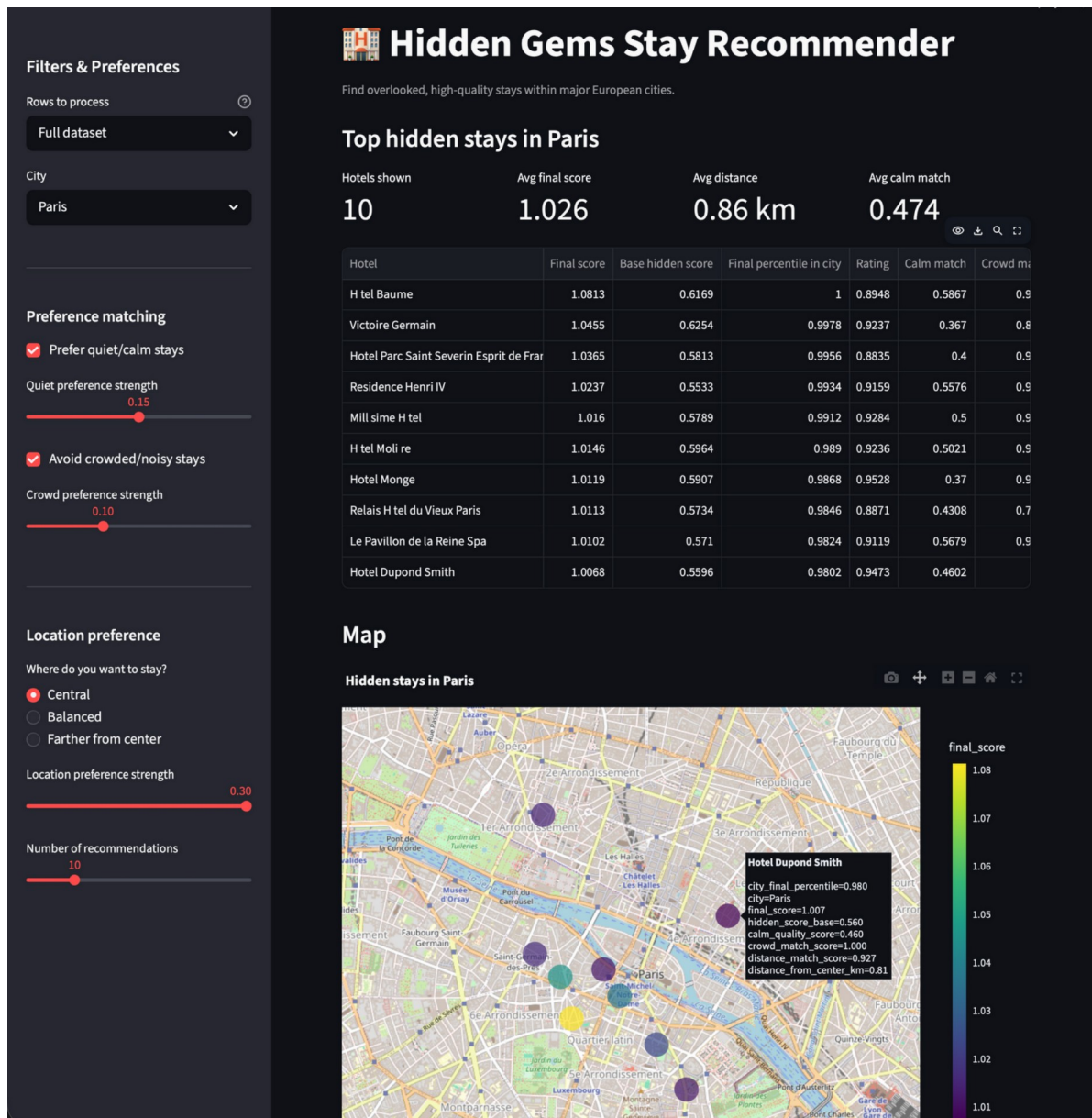


Figure 5. Streamlit application interface. The app includes city selection, preference controls, rankings, map visualization, and explainability metrics.

Figure 5 is included at the end of the results section because it demonstrates the final end-to-end deliverable: the scoring and recommendation logic was converted into an interactive application.

## Findings

- A relatively small group of hotels score higher than the main distribution, making hidden-gem ranking useful for discovery.
- Popularity is an imperfect proxy for quality; normalized review volume does not fully explain hidden-gem score.
- City-level differences matter, so rankings should be evaluated within each city rather than globally.



- Preference controls have a material effect on results, supporting the decision to separate baseline quality from traveler-specific matching.
- Explainability features make recommendations easier to interpret and debug.

## Recommendations

- Use the recommender as a discovery layer for travelers who want high-quality hotels beyond the most popular listings.
- Present users with preference controls for quietness, crowd tolerance, and location style before displaying final rankings.
- Show explainability metrics alongside each recommendation so users can understand whether a hotel matched because of calm quality, crowd profile, or location fit.

## Limitations

- Report figures are based on a 100,000-review development sample, not a full rerun of every notebook figure on all reviews.
- TextBlob is lightweight and interpretable but less context-aware than modern transformer sentiment models.
- Aspect keyword matching can miss nuanced language, sarcasm, multilingual phrasing, or implied sentiment.
- The project uses historical review data and does not include current prices, availability, renovation status, or recent ownership changes.
- The scoring weights are expert-defined rather than learned from observed booking or satisfaction outcomes.

## Future Work

- Replace or supplement TextBlob with transformer-based sentiment models for more context-aware aspect scoring.
- Incorporate current price, seasonality, and availability to improve real-world booking usefulness.
- Add attraction, transit, neighborhood, and safety datasets to strengthen geographic personalization.
- Learn preference weights from user feedback once real usage data is available.
- Evaluate recommendations with user studies or A/B testing rather than relying only on engineered scoring logic.

## Conclusion

The Hidden Gems Hotels Recommender shows how review text, hotel metadata, geographic features, and user preferences can be combined into an interpretable content-based

recommendation system. The project moves beyond popularity-driven ranking by identifying hotels with strong guest experience signals and adjusting results for different traveler profiles. The final Streamlit application demonstrates that the methodology can be delivered as an interactive decision-support tool.